Line Integral

1) Definitions

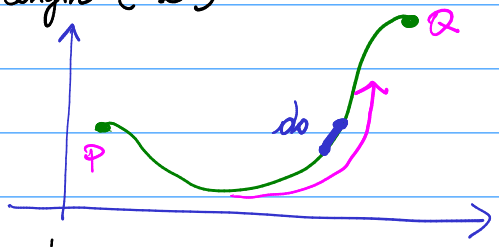
Context

Defn: A line integral is the integration of a multivariate function $f: \mathbb{R}^n \to \mathbb{R}$ $(n = 2, 3)$ over a space curve $\mathcal{C}$ with length s.

Notation: $\int_{\mathcal{C}} f \, ds$

We need:

* $f: \mathbb{R}^n \to \mathbb{R}$ multivariate.
* A space curve $\mathcal{C}$ = Range $(\vec{r})$
  $\vec{r}: [a, b] \to \mathbb{R}^n$ (must coincide)
* s = length $(\mathcal{C})$



$\vec{r}(a) = P$

$\vec{r}(b) = Q.$

$$s = \int_a^b |\vec{r}'(t)| \, dt$$

$$ds^2 = dx^2 + dy^2$$

$$ds = \sqrt{dx^2 + dy^2}$$

$$\begin{cases} x = x(t) \\ y = y(t) \end{cases} \Rightarrow \quad r(t) = \langle x(t), y(t) \rangle$$

$$\Rightarrow \quad \frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

$\sqrt{x^2} = |x|$

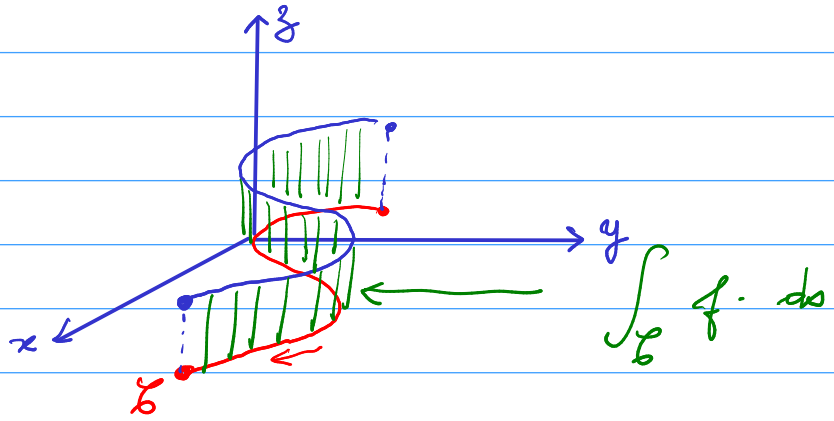
$x \geq 0 \Rightarrow \sqrt{x^2} = x$

$$s = \int_a^b ds \quad \Rightarrow \quad s = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \, dt$$

$$s = \int_a^b |\vec{r}'(t)| \, dt$$

Proposition : If f continuous, $\vec{r}$ continuous and $\vec{r}'(t) \neq 0$, then

$$\int_{\mathcal{C}} f \cdot ds = \int_a^b f(\vec{r}(t)) \cdot |\vec{r}'(t)| \, dt$$

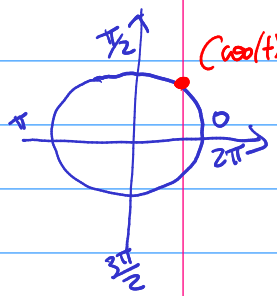


$f : \mathbb{R}^2 \longrightarrow \mathbb{R}$

$f(x, y)$

2) **Examples :** (a) $f(x, y) = x^2 + y^2$

$\mathcal{C} = \{ (x, y) \mid x^2 + y^2 = 1 \, , \, y \geq 0 \}$



$\vec{r}(t) = ?$

$$\begin{cases} x = \cos(t) \\ y = \sin(t) \end{cases} \qquad t \in [0; \pi]$$

$$\vec{r} : [0; \pi] \longrightarrow \mathbb{R}^2$$

$$t \longmapsto \langle \cos(t), \sin(t) \rangle$$

$$\int_{\mathcal{C}} f(x, y) \, ds = \int_0^{\pi} f(\cos(t), \sin(t)) \cdot \sqrt{(-\sin(t))^2 + (\cos(t))^2} \, dt$$

$$= \int_0^{\pi} (\cos^2(t) + \sin^2(t)) \sqrt{\sin^2(t) + \cos^2(t)} \, dt$$

$$= \int_0^{\pi} 1 \cdot dt$$

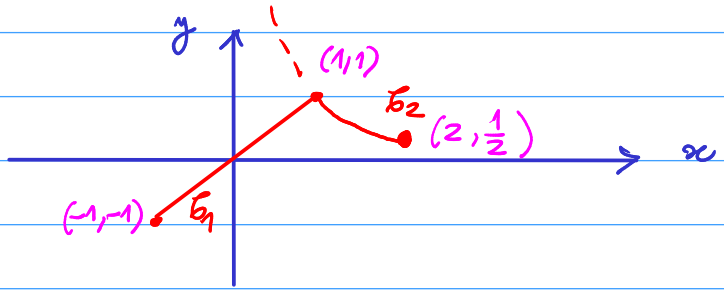
(b) $C_1, C_2 \Rightarrow \int_{C_1 \cup C_2} f \cdot d\sigma = \int_{C_1} f \cdot d\sigma + \int_{C_2} f \cdot d\sigma$

- $f(x,y) = x + y$

$C_1 = \{ (x,y) \mid y = x, \ -1 \le x \le 1 \}$

$C_2 = \{ (x,y) \mid xy = 1, \ 1 \le x \le 2 \}$

$xy = 1 \Leftrightarrow y = \frac{1}{x}$



- $C_1 =$ Range $(\vec{r}_1)$ and $C_2 =$ Range $(\vec{r}_2)$

$\begin{pmatrix} -1 \\ -1 \end{pmatrix} + t \begin{pmatrix} 1-(-1) \\ 1-(-1) \end{pmatrix}$

$P + t\vec{w}$

* $\vec{r}_1(t) = (1-t)\begin{pmatrix} -1 \\ -1 \end{pmatrix} + t\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad t \in [0;1]$

$\vec{r}_1(t) = \begin{pmatrix} (1-t)(-1) + t \\ (1-t)(-1) + t \end{pmatrix} = \langle 2t-1, 2t-1 \rangle$

* $xy = 1 \Leftrightarrow y = \frac{1}{x}$

$\vec{r}_2(t) = \langle t, \frac{1}{t} \rangle, \quad t \in [1,2]$

$\Rightarrow \int_{C_1 \cup C_2} f \cdot d\sigma = \underbrace{\int_0^1 2(2t-1)\sqrt{2^2+2^2}\,dt}_{\text{on } C_1} + \underbrace{\int_1^2 \left(t + \frac{1}{t}\right)\sqrt{1^2 + \left(\frac{-1}{t^2}\right)^2}\,dt}_{\text{on } C_2}$

3) Line integral over C with respect to x and y

Defn: The line integral of $f: D \to \mathbb{R}$ $(D \subseteq \mathbb{R}^n)$ over a curve $C =$ Range $(\vec{r}(t))$ with respect to x

not ds

is given by $\int_C f\,dx = \int_a^b f(\vec{r}(t)) \cdot x'(t)\,dt$.

$x'(t) = \frac{dx}{dt}$ $\left(\int_C f\,dy = \int_a^b f(\vec{r}(t)) \cdot y'(t)\,dt\right)$

